

ATROPOS RESIDUAL OBLIGATION DAG

Final Missed-Atom Inventory — Byte / Token / Parameter / File Level

Continues directly from Tier-1 Hybrid DAG + Full Multi-Tier DAG

Maximum atomization of every omitted item · Outside-the-box mathematical & systems advances · 2026-10-06

This document is the third and final Obligation DAG in the Source Document 7 series. It atomizes every residual item identified by byte-level audit against SD7, SD6 Addenda Parts I & II, the 2050 Scale-Gap Assessment, and the concrete Kotlin/Python reference implementations. It leaves off exactly where the previous two DAGs ended.

Extraction rules locked: maximum atomization, hybrid Residual RESEARCH+IMPL + HOE Track/dependsOn/WIRE, research forces external discovery, impl notes name foreseeable failures and cheapest correct paths.

1. Runtime Byte Ceilings & Throttle Removals (SD7 Diagnostic Core)

SD7-R-001+lift-contextCapBytes [RESIDUAL]

Track: S

dependsOn: []

R: Research dynamic context-window governance used by long-context agent runtimes (Claude Code, Aider, Cursor, Continue.dev) that raise or remove hard byte ceilings without risking OOM on constrained hosts.

I: Mutate AgentContextCollector.kt: replace hardcoded contextCapBytes (4 KB–64 KB) with a profile-driven configurable ceiling that defaults to partition footprint ($\leq 12\,000$ tokens) and can be lifted for factory runs. Foreseeable error: unbounded growth on Termux; keep a hard device-profile ceiling and refuse synthesis if free RAM is below threshold.

WIRE: AgentContextCollector.kt, HostHealthMonitor, PartitionClusterManager

SD7-R-002+lift-maxAdvances [RESIDUAL]

Track: S

dependsOn: [SD7-R-001]

R: Research loop-limit designs that prevent infinite agent cycles while still permitting multi-thousand-step autonomous factory runs.

I: Mutate SelfHostRuntimeRunLimits.kt: replace hardcoded maxAdvances = 15 with a wave-aware limit derived from remaining Φ potential and partition count. Factory mode must allow $\geq 10\,000$ advances. Foreseeable error: silent infinite loop; couple to $\Phi(t+1) < \Phi(t)$ and oscillation prune.

WIRE: SelfHostRuntimeRunLimits.kt, AutonomousWaveOrchestrator, DirectorAgent

SD7-R-003+lift-GlobalByteCeiling [RESIDUAL]

Track: S

dependsOn: [SD7-R-001]

R: Research global memory/context ceilings that scale with available host resources while preserving the clean-room token bound $O(|I|+|a_i|)$.

I: Introduce a GlobalByteCeiling policy atom that is raised for factory mode to permit 50 k–100 k+ LOC ASTs. Still enforce per-worker context convexity. Foreseeable error: single worker saturating host RAM; keep per-partition bound absolute.

WIRE: ExecutionPolicyEngine, AgentContextCollector, HostHealthMonitor

SD7-R-004+deprecate-atoms-md-dump [RESIDUAL]

Track: S

dependsOn: [SD7-T1-003]

R: Research the separation of audit logging from synthesis authority so that atoms written to .atropos/research/atoms.md never become the source of truth for file generation.

I: Refactor or deprecate the path that dumps SpecGraph atoms into .atropos/research/atoms.md while templates still emit files. After Tier-1 template purge, the dump becomes pure audit; synthesis authority is the Wave orchestrator. Foreseeable error: dual sources of truth; ArchitectureComplianceChecker must reject any generation path that reads the dump.

WIRE: AppFactoryRouter, SpecGraphCompiler, ArchitectureComplianceChecker

2. Missing Structural Files & Explicit Code Subsystems

SD7-R-010+ToolchainExecutionDriver [RESIDUAL]

Track: S

dependsOn: [SD7-T1-010]

R: Research sandboxed process harnesses that capture exact stdout/stderr/binary artifacts under timeout, byte and line limits (Bazel sandbox, Bubblewrap, Firejail patterns).

I: Create src/main/kotlin/atropos/core/toolchain/ToolchainExecutionDriver.kt as the dedicated execution harness. Responsibilities: process sandboxing, raw stdout/stderr capture, timeouts, byte limits, clean env. ATOM-SYS-TC-001. Foreseeable error: leaking host env; always start with empty environment and inject only required vars.

WIRE: NativeToolchainRegistry, BoundedProcessRunner, all LanguageToolchainDrivers

SD7-R-011+BehavioralTestSynthesizer [RESIDUAL]

Track: S

dependsOn: [SD7-T1-020, SD7-T1-030]

R: Research dedicated mock-and-invariant test generators that produce suites guaranteed to fail against empty stubs and compile against Wave-0 contracts only.

I: Create src/main/kotlin/atropos/core/factory/BehavioralTestSynthesizer.kt (ATOM-L2-001). Responsibility: mock synthesis + invariant test generation for Wave-1. Must not be collapsed into a generic wave generator. Foreseeable error: tests that pass vacuously; require at least one failing assertion on a stub.

WIRE: ContractLatticeSynthesizer, VerifiedCompletionGate, Wave-1 pipeline

SD7-R-012+mutate-AppProjectSpecParser [RESIDUAL]

Track: S

dependsOn: [SD7-T1-003]

R: Research intent parsers that never short-circuit atom graphs into high-level AppIntent objects that bypass SpecGraph.

I: Mutate or replace AppProjectSpecParser.kt so that parsed intent always flows into SpecGraph atomization; never into a template path. Foreseeable error: residual AppIntent-only branch; delete it.

WIRE: AppFactoryRouter, SpecGraphCompiler, AutonomousWaveOrchestrator

SD7-R-013+DeterministicChecks [RESIDUAL]

Track: S

dependsOn: [SD7-T1-061]

R: Research deterministic AST symbol resolution and package-to-directory alignment checkers used by production monorepos.

I: Ensure DeterministicChecks.kt is wired as a mandatory gate inside VerifiedCompletionGate: AST symbols resolve, packages match directory structures. Foreseeable error: optional check; make it blocking.

WIRE: VerifiedCompletionGate, ArchitectureComplianceChecker, AstSymbolGraph

SD7-R-014+semantic-types-py [RESIDUAL]

Track: B

dependsOn: [SD7-SG-001]

R: Research orthogonal semantic-type taxonomies used by formal requirement engines that prevent dimension collapse.

I: Mutate apps/specgraph-foundry/.../semantic_types.py so that every atom carries a stable semantic type used by Wave-0 dimension filters (INTERFACE, PROTOCOL, SCHEMA, DATA_MODEL). Foreseeable error: free-text dimensions; constrain to closed enum.

WIRE: SpecGraphCompiler, ContractLatticeSynthesizer

SD7-R-015+requirement-quality-py [RESIDUAL]

Track: B

dependsOn: [SD7-SG-001]

R: Research lexical vagueness detectors (VAGUE_TERMS) and requirement quality scoring used by aerospace/medical requirements tools.

I: Mutate requirement_quality.py to flag VAGUE_TERMS and reject or escalate under-specified atoms before Wave-0. Foreseeable error: silent acceptance of vague goals; route to ILSS epistemic bounding.

WIRE: SpecGraphCompiler, ILSS pipeline, FactoryClarificationRequest

SD7-R-016+unresolved-tracker-py [RESIDUAL]

Track: B

dependsOn: [SD7-SG-001]

R: Research ungrounded-requirement trackers that keep a durable ledger of atoms lacking provenance.

I: Mutate unresolved_tracker.py so that every ungrounded candidate is recorded and blocks Phase-100 until resolved or explicitly waived. Foreseeable error: lost unresolved set across restarts; store in CAS.

WIRE: SpecGraphCompiler, ResidualObligationLedger, CAS

SD7-R-017+Sd5HumanIntentInvariants-ILSS-bridge [RESIDUAL]

Track: S

dependsOn: [SD7-2050-ILSS-01, SD7-2050-ILSS-03]

R: Research authority lattices that keep human prompt primacy (HIA-001–HIA-008) while still permitting adversarial goal expansion.

I: Bridge Sd5HumanIntentInvariants.kt into the ILSS pipeline so that rank(research) < rank(human_intent) is enforced even when ¬S is generated. Foreseeable error: ILSS silently redefining objectives; SourceAuthorityLaw must reject.

WIRE: Sd5HumanIntentInvariants.kt, ILSS, SourceAuthorityLaw, FactoryClarificationRequest

SD7-R-018+TreeSitterGrammarBridge [RESIDUAL]

Track: S

dependsOn: [SD7-T1-071]

R: Research production Tree-sitter grammar bridges that expose a uniform AST walk API across Kotlin, TypeScript, Python, Rust, Go.

I: Create the concrete TreeSitterGrammarBridge class mandated by SD6 EXTIRPATION_01. All AST stub/hollow scans and call-graph walks must go through this bridge. Foreseeable error: language-specific ad-hoc parsers; single bridge only.

WIRE: ArchitectureComplianceChecker, call-graph reachability, AST-stub scanner

3. Low-Level OS Sandboxing & Process Jailing

SD7-R-020+chroot-unshare-jail [RESIDUAL]

Track: B

dependsOn: [SD7-R-010, SD7-T3-001]

R: Research Linux namespace isolation (unshare, chroot, bubblewrap, Firejail, landlock) that provides READ_ONLY_ROOT + WRITEABLE_WORKTREE with minimal environment.

I: Execute every compiler/toolchain inside chroot/unshare or container sandbox. Filesystem policy: READ_ONLY_ROOT_DIRECTORY (/) + only the isolated worktree directory mounted writeable. Secrets stripped. Foreseeable error: host filesystem escape; fail closed on any path outside the worktree.

WIRE: ToolchainExecutionDriver, BoundedProcessRunner, IsolatedWorktreeService

4. Delivery Platforms — CONTAINER & LOCAL_DIST

SD7-R-030+TargetPlatform-CONTAINER [RESIDUAL]

Track: B

dependsOn: [SD7-T4-010]

R: Research deterministic OCI/container image assembly from verified build artifacts (Buildah, kaniko, Docker BuildKit rootless).

I: Add TargetPlatform.CONTAINER. Produce bit-verified OCI image with SHA-256 manifest. Foreseeable error: non-reproducible layers; pin base images by digest.

WIRE: CloudDeploymentAdapter, AutonomousDeliveryEngine, TargetPlatform enum

SD7-R-031+TargetPlatform-LOCAL_DIST [RESIDUAL]

Track: B

dependsOn: [SD7-T4-010]

R: Research local distribution packaging that emits ELF/APK/JAR + SHA-256 proof manifests to a dist/ directory.

I: Add TargetPlatform.LOCAL_DIST (or LOCAL_BINARY). Package to disk with bit-level SHA-256 proof manifests. Acceptance = valid signed/bit-verified binary on disk. Foreseeable error: missing proof file; VerifiedCompletionGate refuses.

WIRE: CloudDeploymentAdapter, AutonomousDeliveryEngine, TargetPlatform enum, IntegrityProofs

SD7-R-032+Google-Drive-CAS-cold-tier [RESIDUAL]

Track: B

dependsOn: [SD7-T2-020]

R: Research Google Drive as an additional content-addressed cold-tier backend alongside R2.

I: Storage Constitution COLD tier: REMOTE_R2_OR_DRIVE_CAS_BLOBS. Implement Drive CAS adapter for cold evidence and build products. Foreseeable error: Drive API rate limits; treat as best-effort with R2 primary.

WIRE: CAS, Custodian, Storage Constitution

SD7-R-033+COMPILER_TEMP_JARS-eviction [RESIDUAL]**Track:** B**dependsOn:** [SD7-T2-020]**R:** Research automatic eviction of compiler temporary JARs and intermediate build products.**I:** EXPIRE retention tier must explicitly include COMPILER_TEMP_JARS. Custodian sweeps them after every wave. Foreseeable error: leftover temp JARs filling flash; include in mark-and-sweep.**WIRE:** Custodian, Storage Constitution, IsolatedWorktreeService

5. Schema-Driven Type Serialization Integrity (INVARIANT_04)

SD7-R-040+runtime-codec-synthesis [RESIDUAL]**Track:** S**dependsOn:** [SD7-T1-020, SD7-T4-001]**R:** Research mandatory round-trip serializer generation (Zod, Pydantic, Serde, kotlinx.serialization) that enforces ingress/egress validation, unknown-field rejection and fail-closed overflow.**I:** Wave-0 or Wave-3 must synthesize bidirectional runtime codecs for every boundary type. Rules: all ingress/egress validated at runtime boundary; unknown fields rejected; payload overflow fails closed at parser entry. Foreseeable error: optional codecs; make them mandatory for any public boundary.**WIRE:** ContractLatticeSynthesizer, DependencyInjectionAssembler, ArchitectureComplianceChecker

6. Mathematical & Information-Theoretic Operators (AACS-2036 Part 0)

SD7-R-050+memory-fold-operator [RESIDUAL]**Track:** S**dependsOn:** [SD7-T1-040, SD7-T1-018]**R:** Research internal state-transition operators for isolated agents that fold successive atoms without reloading the full codebase (memory-fold, recurrent state machines, linear attention folds).**I:** Implement $\sigma_{t+1} = \delta(\sigma_t, a_{\{t\}}, I)$ inside each partition worker so that topological atom sequence is processed with constant context. Foreseeable error: context growth; enforce that $|\sigma|$ stays bounded by partition footprint.**WIRE:** DecoupledWorkerSynthesizer, PartitionClusterManager**SD7-R-051+context-convexity-bound** [RESIDUAL]**Track:** S**dependsOn:** [SD7-R-050]**R:** Research formal proofs that context remains $O(|I| + |a_i|)$ under ancestral interface inclusion.**I:** Enforce $X(\tau_i) = I \cup \{a_i\} \cup \text{Ancestors}(a_i, I)$. Any prompt that exceeds this bound is rejected before provider dispatch. Foreseeable error: accidental inclusion of sibling code; scrub and assert.**WIRE:** AgentPromptContract, DecoupledWorkerSynthesizer, ArchitectureComplianceChecker**SD7-R-052+zero-entropy-constraint** [RESIDUAL]**Track:** S**dependsOn:** [SD7-T1-060]**R:** Research operational definitions of $\text{Entropy}(C \mid S, I) = 0$ that can be approximated by the conjunction of all verification oracles.**I:** The synthesis operator Φ must satisfy $\text{Entropy}(C \mid S, I) = 0$ as the formal objective. Approximated by the product of all verifiers = 1 and the Zero-Entropy Proof Certificate. Foreseeable error: treating it as rhetoric; wire it as a blocking predicate in Phase-100.**WIRE:** VerifiedCompletionGate, Zero-Entropy Proof Certificate**SD7-R-053+edge-qualified-decoupling** [RESIDUAL]**Track:** S**dependsOn:** [SD7-T1-021]**R:** Research mutual-information nullification conditioned on the absence of a dependency edge.**I:** $I(f_j; f_k \mid I) = 0 \ \forall j \neq k$ such that $(v_j, v_k) \notin E_A$. The edge condition must be checked by the partitioner; workers that share an edge may see ancestral contracts only. Foreseeable error: ignoring the edge qualifier; enforce it in Ψ .**WIRE:** PartitionClusterManager, AgentPromptContract**SD7-R-054+verifier-ensemble-product** [RESIDUAL]**Track:** S**dependsOn:** [SD7-T1-061]**R:** Research product calculi over deterministic verifier ensembles that refuse partial success.**I:** $\prod \{v \in V\} v(\Phi(P_r, I)) = 1$ where $V = \{v_{\text{syntax}}, v_{\text{type}}, v_{\text{property}}, v_{\text{boundary}}\}$. Any single 0 fails the partition. Foreseeable error: soft-AND; implement hard product.**WIRE:** VerifiedCompletionGate, GovernedCompileGate**SD7-R-055+error-gradient-diff-subspace** [RESIDUAL]**Track:** S**dependsOn:** [SD7-T1-041]**R:** Research repair loops that restrict mutations to the territory of the failing partition (diff-subspace restriction).**I:** $\nabla E = \text{Diagnostics}(v)$ and $\Delta f \subseteq \text{Territory}(P_r)$. Repair must never touch files outside the partition territory. Foreseeable error: global search-and-replace; enforce territory lock.**WIRE:** CompilerDiagnosticRepairLoop, TerritoryGrant, IsolatedWorktreeService**SD7-R-056+swarm-acceptance-volume** [RESIDUAL]**Track:** S**dependsOn:** [SD7-T2-012]**R:** Research four-variable acceptance volume operators used by large agent systems.**I:** $V_{\text{accepted}} = \min(V_{\text{generated}}, V_{\text{integrated}}, V_{\text{verified}}, V_{\text{evidenced}})$. Wire into Director metrics. Foreseeable error: counting generated lines as accepted; always take the min.**WIRE:** DirectorAgent, EvidenceCollector, VerifiedCompletionGate**SD7-R-057+phase100-acceptance-velocity** [RESIDUAL]**Track:** S**dependsOn:** [SD7-R-056, SD7-T2-023]

R: Research Phase-100 predicates and acceptance-velocity metrics that declare a multi-week run complete.

I: Phase 100 = $\wedge(\text{MandatoryPredicates}) \wedge \text{RuntimeAcceptance} \wedge \text{NegativePathTests} \wedge \text{EvidenceComplete} \wedge \neg \text{FakeSuccess} \wedge \neg \text{CriticalOrphan}$. $\text{AcceptanceVelocity} = \Delta \text{VerifiedAcceptancePredicates} / \Delta \text{Days}$. Both must be computed and stored in the governance ledger. Foreseeable error: declaring success on compile alone; require full conjunction.

WIRE: VerifiedCompletionGate, DirectorAgent, GovernanceLedger, EvidenceCollector

SD7-R-058+partition-scaling-complexity [RESIDUAL]

Track: S

dependsOn: [SD7-T1-018]

R: Research complexity proofs that synthesized partitions scale $O(N/K)$ independent of total system lines.

I: Document and assert that partition work is $O(N/K)$. Instrument and fail CI if measured complexity exceeds the bound by a constant factor. Foreseeable error: quadratic hidden costs; measure end-to-end.

WIRE: PartitionClusterManager, DirectorAgent

7. Concrete Kotlin Data Contracts, Enums & HTTP Wire Specs

SD7-R-060+ExtractionAtom-model [RESIDUAL]

Track: S

dependsOn: []

R: Research minimal complete atom models that carry id, statement, dimension, dependencies, estimatedTokens, targetPath.

I: Define data class ExtractionAtom(id, statement, dimension, dependencies, estimatedTokens, targetPath) as the canonical atom type used by every downstream component. Foreseeable error: ad-hoc maps; single sealed model only.

WIRE: SpecGraphCompiler, PartitionClusterManager, DecoupledWorkerSynthesizer

SD7-R-061+SynthesisPartition-model [RESIDUAL]

Track: S

dependsOn: [SD7-R-060]

R: Research partition models that carry partitionId, atoms, targetFiles, requiredContracts, dependencies, totalEstimatedTokens.

I: Define data class SynthesisPartition(...) exactly. All Ψ output must be this type. Foreseeable error: lossy intermediate representations; preserve full fidelity.

WIRE: PartitionClusterManager, DecoupledWorkerSynthesizer, AutonomousWaveOrchestrator

SD7-R-062+RepairLoopResult-ToolchainResult [RESIDUAL]

Track: S

dependsOn: [SD7-T1-041]

R: Research structured result types that carry success, attempts, diagnostics, exitCode, and full diagnostic maps (file, line, column, error_code, trace).

I: Define RepairLoopResult(success, attempts, diagnostics) and ToolchainResult(passed, exitCode, diagnostics). Diagnostic map must include error_code and trace. Return status CONVERGED on success. Foreseeable error: stringly diagnostics; structured only.

WIRE: CompilerDiagnosticRepairLoop, ToolchainExecutionDriver, VerifiedCompletionGate

SD7-R-063+TargetPlatform-enum-complete [RESIDUAL]

Track: S

dependsOn: [SD7-R-030, SD7-R-031]

R: Research exhaustive target-platform enums that include LOCAL_BINARY, ANDROID_APK, VERCEL, NETLIFY, CONTAINER, and any future LOCAL_DIST alias.

I: enum class TargetPlatform { LOCAL_BINARY, ANDROID_APK, VERCEL, NETLIFY, CONTAINER }. All delivery drivers must pattern-match exhaustively. Foreseeable error: silent fall-through; use sealed when.

WIRE: AutonomousWaveOrchestrator, CloudDeploymentAdapter, AutonomousDeliveryEngine

SD7-R-064+BuildExecutionReport-sealed [RESIDUAL]

Track: S

dependsOn: [SD7-T1-060]

R: Research sealed report hierarchies with explicit typed phase failure constants.

I: BuildExecutionReport = Success(totalFilesGenerated, artifactOnDisk, deploymentUrl) | Halted(phase, diagnostics). Phase strings must be the closed set: WAVE_0_FAILURE, WAVE_3_SYNTHESIS_FAILURE, WAVE_4_ASSEMBLY_FAILURE, GLOBAL_COMPILE_FAILED, GLOBAL_TEST_FAILED, PACKAGING_FAILED. Foreseeable error: free-text phases; constrain.

WIRE: AutonomousWaveOrchestrator, VerifiedCompletionGate

SD7-R-065+Wave0-dimension-filter [RESIDUAL]

Track: S

dependsOn: [SD7-R-014, SD7-T1-020]

R: Research dimension filters that select only INTERFACE/PROTOCOL/SCHEMA/DATA_MODEL atoms for Wave-0.

I: Wave-0 contracts = atoms.filter { it.dimension in setOf("INTERFACE", "PROTOCOL", "SCHEMA", "DATA_MODEL") }. Enforce closed set. Foreseeable error: leaking implementation atoms into Wave-0; filter strictly.

WIRE: ContractLatticeSynthesizer, SpecGraphCompiler

SD7-R-066+directed-RepairNode-on-edge [RESIDUAL]

Track: S

dependsOn: [SD7-T1-060, SD7-R-055]

R: Research directed repair nodes that target a single failing integration edge instead of aborting the entire build.

I: If an integration edge breaks in Wave-4, create a directed RepairNode in the DAG targeted specifically at the failing edge; do not abort the whole build. Foreseeable error: global abort; schedule edge-local repair.

WIRE: AutonomousWaveOrchestrator, CompilerDiagnosticRepairLoop, DirectorAgent

SD7-R-067+Vercel-REST-wire-contract [RESIDUAL]

Track: B

dependsOn: [SD7-T4-011]

R: Research the exact Vercel Deployments API contract used by production deployers.

I: File upload: POST https://api.vercel.com/v2/files?teamId=... with header x-vercel-digest: <sha1>. Deployment manifest: POST https://api.vercel.com/v13/deployments?teamId=... with projectSettings.framework = null. HTTP client timeout Duration.ofSeconds(30). Foreseeable error: wrong endpoint version; pin the documented paths.

WIRE: VercelDeploymentDriver, SecretSinkMatrix

8. Hardcoded Numerical Limits, Timeouts & Benchmark Criteria

SD7-R-070+AndroidSdkDriver-concrete-thresholds [RESIDUAL]

Track: B

dependsOn: [SD7-T1-017]

R: Research production Android build timeouts and buffer ceilings used by CI systems.

I: AndroidSdkDriver: compile/test timeoutMillis = 600_000 (10 min), assemble timeoutMillis = 900_000 (15 min), maxOutputBytes = 4*1024*1024 (4 MB), maxOutputLines = 5000. Tasks: compileDebugSources --no-daemon --stacktrace, testDebugUnitTest --no-daemon, assembleDebug --no-daemon. APK path: app/build/outputs/apk/debug.

Foreseeable error: silent truncation; enforce the ceilings.

WIRE: AndroidSdkDriver, ToolchainExecutionDriver, BoundedProcessRunner

SD7-R-071+Kahn-token-floor [RESIDUAL]

Track: S

dependsOn: [SD7-T1-018]

R: Research minimum token-weight floors that prevent zero-cost atoms from collapsing partitions.

I: atom.estimatedTokens.coerceAtLeast(250). Apply inside Ψ . Foreseeable error: zero-weight atoms; clamp.

WIRE: PartitionClusterManager

SD7-R-072+chaos-error-rate-ceiling [RESIDUAL]

Track: B

dependsOn: [SD7-2050-FMC-02]

R: Research quantitative chaos acceptance thresholds used by production resilience systems.

I: During Digital Twin chaos: error rate must remain $< 10^{-6}$; zero unhandled promise or background thread rejections. Foreseeable error: qualitative 'survived'; require the numeric bound.

WIRE: Chaos Digital Twin, VerifiedCompletionGate, EvidenceCollector

SD7-R-073+partition-10k-node-benchmark [RESIDUAL]

Track: B

dependsOn: [SD7-T1-018]

R: Research synthetic DAG generators that produce 10 000-node acyclic graphs for partitioner validation.

I: ATOM-SYS-PL-001 verification: partitioner must pass against a synthetic 10 000-node dependency DAG with zero cycles. Foreseeable error: only testing small graphs; make the 10 k-node case a CI gate.

WIRE: PartitionClusterManager, CI job

SD7-R-074+ingestion-5k-page-512MB [RESIDUAL]

Track: B

dependsOn: [SD7-SG-001]

R: Research streaming ingestion under hard RAM ceilings.

I: ATOM-ING-001 verification: process a 5 000-page specification stream under a strict 512 MB RAM ceiling. Foreseeable error: materializing the whole stream; stream only.

WIRE: SpecGraphCompiler, CI job

SD7-R-075+schema-ephemeral-sqlite [RESIDUAL]

Track: B

dependsOn: [SD7-T4-001]

R: Research ephemeral in-memory SQLite validation of generated DDL and RLS before migration emission.

I: ATOM-L1-003 verification: run generated DDL/RLS against an ephemeral in-memory SQLite parser before emitting migrations. Foreseeable error: emitting unvalidated SQL; always validate first.

WIRE: StorageSchemaCompiler, SupabaseSchemaDriver, VerifiedCompletionGate

9. Additional Verification Domains, Governance & Formal Engines

SD7-R-080+WCAG-STRIDE-ILSS [RESIDUAL]

Track: S

dependsOn: [SD7-2050-ILSS-02]

R: Research automatic grafting of WCAG 2.2 AA accessibility contracts and STRIDE threat models from domain corpora.

I: ILSS DLOI extrapolation must include WCAG 2.2 AA and STRIDE as mandatory non-functional lattices when the domain implies UI or networked attack surface.

Foreseeable error: optional; make them first-class Wave-0 atoms for relevant domains.

WIRE: ILSS, DloiService, ContractLatticeSynthesizer

SD7-R-081+Dafny-JavaPathfinder [RESIDUAL]

Track: B

dependsOn: [SD7-2050-NSV-01, SD7-2050-NSV-02]

R: Research integration of Dafny and Java Pathfinder as additional formal verification oracles alongside Kani/Creusot/icontract.

I: Neuro-symbolic SMT path must support Dafny and Java Pathfinder for languages/toolchains where they are the strongest available. Foreseeable error: single-oracle lock-in; keep the ensemble open.

WIRE: LanguageToolchainDriver, SMT verification path, VerifiedCompletionGate

SD7-R-082+sequential-Director-merge-arbitration [RESIDUAL]

Track: S

dependsOn: [SD7-T2-010, SD7-T1-040]

R: Research sequential merge arbitration locks that prevent concurrent worktree merges.

I: SUBAGENTS_NEVER_MERGE_DIRECTLY; ALL_MERGES_ARBITRATED_SEQUENTIALLY_BY_DIRECTOR_UNDER_VerifiedCompletionGate. Implement a Director-held merge lock. Foreseeable error: parallel merges; serialize.

WIRE: DirectorAgent, IsolatedWorktreeService, VerifiedCompletionGate

SD7-R-083+UTF8-key-decoder-ring-buffer [RESIDUAL]

Track: B

dependsOn: [SD7-T3-005]

R: Research searchable ring buffers and raw UTF-8 key decoders for terminal input.

I: CURSOR_KEYBOARD = VT100_ESCAPES + UTF8_RAW_KEY_DECODER + SEARCHABLE_RING_BUFFER. Complete the terminal input substrate. Foreseeable error: incomplete key handling; implement the full triple.

WIRE: CLI input loop, Terminal Input Defense

SD7-R-084+termux-clipboard-set [RESIDUAL]

Track: B

dependsOn: [SD7-T3-006]

R: Research Termux native clipboard binary binding.

I: OSC-52 fallback cascade must explicitly include termux-clipboard-set as the first Android/Termux target. Foreseeable error: missing Termux path; place it first in the cascade.

WIRE: OSC-52 clipboard bridge, SecretSinkMatrix

SD7-R-085+zero-reward-fraud-penalty-ledger [RESIDUAL]

Track: S

dependsOn: [SD7-T1-071]

R: Research zero-reward penalty ledgers that permanently record fraudulent synthesis attempts.

I: When AST stub scanner rejects code as fraudulent, log a ZERO_REWARD_PENALTY entry into the governance ledger. Future routing may down-rank the responsible provider. Foreseeable error: silent reject; always ledger.

WIRE: ArchitectureComplianceChecker, GovernanceLedger, RoutePolicy

SD7-R-086+true-moat-governance-ledger-amendment-chain [RESIDUAL]

Track: S

dependsOn: [SD7-T2-001, SD7-T2-025]

R: Research the true moat components: stateful evidence substrate, CAS-addressable records, restart continuity hashes, governance ledger, amendment chain.

I: TRUE_MOAT = STATEFUL_EVIDENCE_SUBSTRATE + CAS_ADDRESSABLE_RECORDS + RESTART_CONTINUITY_HASHES + GOVERNANCE_LEDGER + AMENDMENT_CHAIN. Implement GovernanceLedger and AmendmentChain as first-class durable stores. Foreseeable error: treating moat as rhetoric; wire every component.

WIRE: GovernanceLedger (new), AmendmentChain (new), CAS, EvidenceCollector, RestartCoordinator

10. Outside-the-Box Mathematical & Systems Advances

Atoms that push beyond Source Document 7 using techniques from formal methods, distributed systems, baseball/football analytics-style measurement, and CLI design that the original documents have not yet incorporated.

SD7-R-090+information-bottleneck-partitioning [AMENDMENT-CANDIDATE]

Track: S

dependsOn: [SD7-T1-018, SD7-R-051]

R: Research the Information Bottleneck method (Tishby) applied to graph partitioning so that each partition maximizes relevant mutual information with its contracts while minimizing leakage to sibling partitions.

I: Replace pure greedy domain-path clustering with an IB-inspired objective that explicitly minimizes $I(P_r; P_s)$ while preserving predictive power for the contracts.

Foreseeable error: intractable optimization; use variational approximation with hard territory constraints.

WIRE: PartitionClusterManager

SD7-R-091+differential-privacy-for-prompts [AMENDMENT-CANDIDATE]

Track: S

dependsOn: [SD7-T1-021]

R: Research differential-privacy noise injection into prompts that still preserve exact interface contracts (useful when multiple untrusted providers see the same Wave-0 lattice).

I: When a contract lattice is sent to more than one provider family, apply calibrated noise to non-semantic fields while keeping type signatures bit-identical. Foreseeable error: breaking typecheck; never noise the actual signatures.

WIRE: AgentPromptContract, ProviderCascadeRouter

SD7-R-092+sabermetrics-acceptance-velocity [AMENDMENT-CANDIDATE]

Track: S

dependsOn: [SD7-R-057]

R: Research sabermetrics-style rate statistics (OPS, WAR analogues) for autonomous software factories that normalize AcceptanceVelocity by partition difficulty and provider cost.

I: Define $\text{FactoryWAR} = (\Delta \text{VerifiedPredicates} - \text{expected}) / \text{cost-normalized replacement level}$. Track per-provider and per-domain. Foreseeable error: vanity metrics; keep the definition anchored to Phase-100 predicates.

WIRE: DirectorAgent, GovernanceLedger, QuotaLedger

SD7-R-093+cli-progress-as-pure-stderr-protocol [AMENDMENT-CANDIDATE]

Track: B

dependsOn: [SD7-T3-004]

R: Research pure-stderr progress protocols used by modern CLIs (cargo, npm, docker) that never contaminate stdout machine output.

I: All progress, spinners, and human telemetry must be pure stderr with an explicit --json-progress alternative that emits NDJSON on stderr only. Stdout remains pure artifacts or machine JSON. Foreseeable error: progress on stdout; forbid by architecture test.

WIRE: CLI TTY discipline, all diagnostic emitters

SD7-R-094+formal-proof-carrying-code-bundle [AMENDMENT-CANDIDATE]

Track: S

dependsOn: [SD7-R-052, SD7-2050-LHD-04]

R: Research proof-carrying code (Necula) and modern PCC variants that ship machine-checkable proofs alongside binaries so that a third party can validate without re-running the LLM.

I: Extend the Zero-Entropy Proof Certificate into a full proof-carrying bundle: SpecHash + NegativeSpecHash + SMT proofs + TLA receipts + mutation-kill score + chaos receipt + toolchain binary hash + optional PCC certificate. Foreseeable error: missing fields; VerifiedCompletionGate refuses promotion until complete.

WIRE: Zero-Entropy Proof Certificate, IntegrityProofs, VerifiedCompletionGate

SD7-R-095+online-convex-optimization-for-routing [AMENDMENT-CANDIDATE]

Track: S

dependsOn: [SD7-T2-024, SD7-ADV-004]

R: Research online convex optimization / multi-armed bandit algorithms that continuously re-rank providers under non-stationary quality and cost.

I: Replace static RoutePolicy rankings with an OCO/bandit update after every verified atom. Regret bounds must be logged. Foreseeable error: catastrophic forgetting; keep a slow-moving prior.

11. Final Residual Completeness Gate

SD7-R-099+residual-completeness-gate [RESIDUAL]

Track: S

dependsOn: [ALL residual atoms above]

R: Research orphan and residual-atom detection that proves every file, constant, equation and invariant named in Source Document 7 and the SD6 addenda now has a corresponding atom.

I: atropos doctor --residual-completeness must pass. It cross-checks the three DAG documents against the full SD7 + SD6 inventory and fails on any remaining omission. Foreseeable error: false green from partial matches; require exact file-path and equation coverage.

WIRE: CI job, atropos doctor, ArchitectureComplianceChecker

12. Ordering Relative to Prior DAGs

Tier-1 Hybrid DAG (template purge + Wave factory + 2050 neuro-symbolic) must be green first.
Full Multi-Tier DAG (Tiers 2/3/4 + SpecGraph expansions + advanced research) must be green second.
This Residual DAG is the third and final layer. SD7-R-099 is the absolute last gate before the system is declared complete with respect to Source Document 7 and its addenda.

13. Non-Negotiable Invariants (Restated for Residual Layer)

Entropy(C | S, I) = 0
 $I(f_j; f_k | I) = 0 \quad \forall (v_j, v_k) \notin E_A$
 $X(\tau_i) = I \cup \{a_i\} \cup \text{Ancestors}(a_i, I)$
 $\sigma_{\{t+1\}} = \delta(\sigma_t, a_{\{t,t\}}, I)$
 $\prod v(\Phi(P_r, I)) = 1$
 $\Delta f \subseteq \text{Territory}(P_r)$
 $V_{\text{accepted}} = \min(V_{\text{generated}}, V_{\text{integrated}}, V_{\text{verified}}, V_{\text{evidenced}})$
Phase 100 full conjunction
False-Green Tolerance = 0.0000 %
Air-Gap for pure synthesis
Exact-hash restart continuity

END OF RESIDUAL OBLIGATION DAG — Final Missed-Atom Inventory

Third and final document in the Source Doc 7 series · Leaves off from Tier-1 + Multi-Tier DAGs · Byte-level completeness · Outside-the-box advances included · 2026-10-06